

# CO4 - AT3 Correlation and Pattern Analysis

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**Title: Study time has almost no relationship with academic performance**

## **Aim :**

To analyze the relationship between study time and academic performance using correlation analysis, subgroup analysis, outlier detection, interaction effects, and visualization techniques.

## **Tools Used :**

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

## **Procedure :**

- Create or load a dataset containing study time, academic marks, and previous academic preparation.
- Calculate the overall correlation between study time and marks.
- Divide students into groups based on their previous preparation level.

- Calculate correlations separately for well-prepared, moderately prepared, and poorly prepared students.
- Visualize the relationship using scatter plots.
- Identify possible outliers, especially students who study for extremely long hours but receive low marks.
- Perform regression analysis to examine the effect of study time and previous preparation.
- Include an interaction term to determine whether the effect of study time differs between preparation groups.
- Compare the overall results with subgroup-level results before making a conclusion.

### Code :

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Create sample student dataset
np.random.seed(42)

n = 2000

preparation = np.random.choice(
    ["Well Prepared", "Moderately Prepared", "Poorly Prepared"],
    n,
    p=[0.3, 0.5, 0.2]
)
```

```
study_time = np.random.uniform(1, 12, n)
```

```
marks = []
```

```
for prep, study in zip(preparation, study_time):
```

```
    if prep == "Well Prepared":
```

```
        mark = 55 + study * 3 + np.random.normal(0, 8)
```

```
    elif prep == "Moderately Prepared":
```

```
        mark = 45 + study * 1.8 + np.random.normal(0, 10)
```

```
    else:
```

```
        mark = 40 + study * 0.3 + np.random.normal(0, 12)
```

```
    marks.append(mark)
```

```
# Create DataFrame
```

```
df = pd.DataFrame({  
    "Preparation": preparation,  
    "Study_Time": study_time,  
    "Marks": marks  
})
```

```
# Add some extreme outliers
```

```
outliers = pd.DataFrame({
    "Preparation": ["Poorly Prepared"] * 20,
    "Study_Time": np.random.uniform(15, 20, 20),
    "Marks": np.random.uniform(10, 35, 20)
})
```

```
df = pd.concat([df, outliers], ignore_index=True)
```

```
# Overall correlation
```

```
overall_corr = df["Study_Time"].corr(df["Marks"])
```

```
print("Overall Correlation:")
```

```
print(overall_corr)
```

```
# Correlation within each subgroup
```

```
group_corr = df.groupby("Preparation").apply(
    lambda x: x["Study_Time"].corr(x["Marks"])
)
```

```
print("\nCorrelation by Preparation Level:")
```

```
print(group_corr)
```

```
# Scatter plot
```

```
plt.figure(figsize=(10, 6))
```

```
sns.scatterplot(
```

```
data=df,  
x="Study_Time",  
y="Marks",  
hue="Preparation"  
)
```

```
plt.title("Study Time vs Academic Performance")  
plt.xlabel("Study Time (Hours)")  
plt.ylabel("Academic Marks")  
plt.grid()
```

```
plt.show()
```

```
# Box plot for outlier analysis  
plt.figure(figsize=(8, 5))
```

```
sns.boxplot(  
    data=df,  
    x="Preparation",  
    y="Marks"  
)
```

```
plt.title("Marks Distribution by Preparation Level")
```

```
plt.show()
```

```
# Correlation heatmap
numeric_df = df[["Study_Time", "Marks"]]

plt.figure(figsize=(6, 4))

sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap="coolwarm")
plt.title("Correlation Between Study Time and Marks")

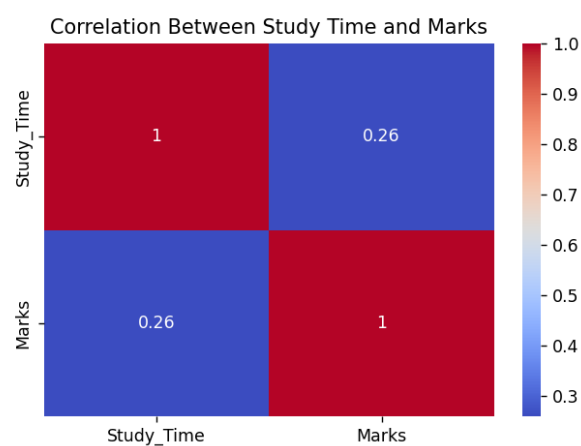
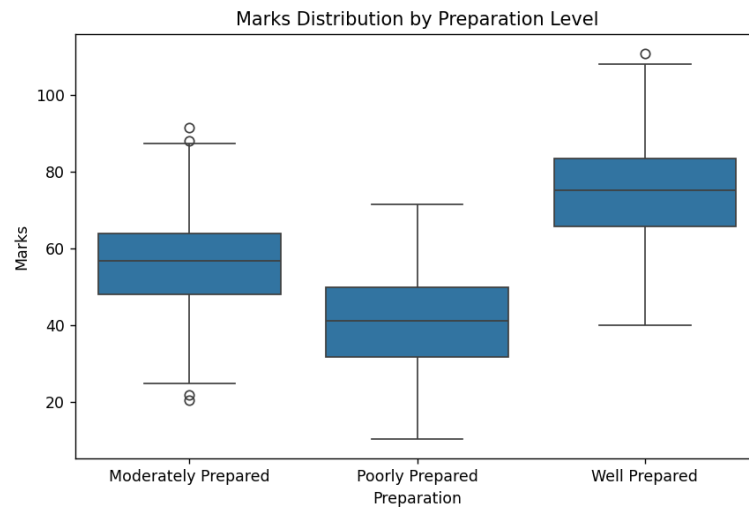
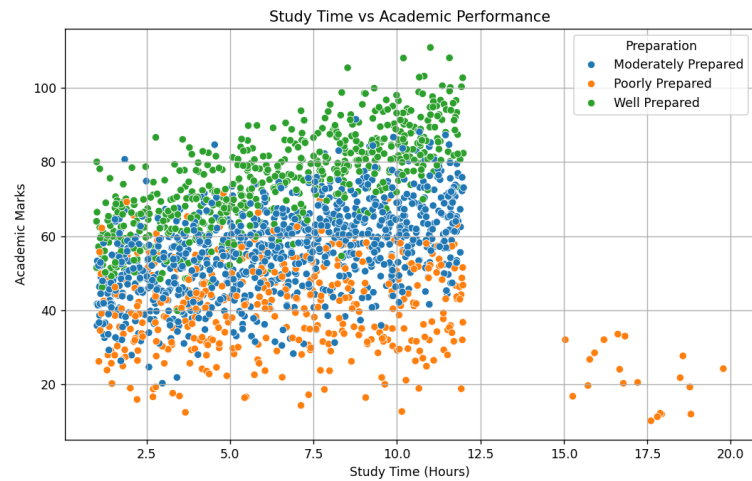
plt.show()
```

### **Output :**

The analysis calculates the overall correlation between study time and academic marks. The overall correlation may appear weak because all students are combined into a single dataset.

However, when students are divided based on their previous academic preparation, different relationships can be observed:

- Well-prepared students show a strong positive relationship.
- Moderately prepared students show a moderate positive relationship.
- Poorly prepared students show a weak relationship.
- Students with extremely high study hours and very low marks appear as possible outliers.



The scatter plot helps visualize these hidden patterns, while the box plot helps identify unusual observations.

## **Key Insights :**

The overall correlation alone does not provide a complete understanding of the relationship between study time and academic performance.

Previous academic preparation can influence how effectively students convert study time into academic performance. This creates an interaction effect, where the impact of study time differs across groups.

A small number of students studying extremely long hours but receiving low marks can act as outliers and reduce or distort the overall correlation.

Therefore, subgroup analysis is important because combining all students may hide meaningful relationships present within individual groups.

## **Conclusion :**

The analysis shows that a single overall correlation is not sufficient to conclude that study time has almost no relationship with academic performance. Subgroup analysis reveals that the relationship can vary depending on previous academic preparation.

Outliers and interaction effects should also be examined before making a final conclusion. Scatter plots, subgroup correlations, outlier detection, and regression analysis provide a more reliable understanding of the relationship between study time and academic performance.



# **Title: Advertising Expenditure and Sales Analysis**

## **Aim :**

To analyze the relationship between advertising expenditure and sales using correlation analysis, time-based analysis, trend detection, seasonality analysis, subgroup comparison, and statistical visualization.

## **Tools Used**

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

## **Procedure :**

- Create or load five years of monthly advertising and sales data.
- Calculate the overall correlation between advertising expenditure and sales.
- Visualize advertising expenditure and sales over time.
- Identify long-term trends in both variables.
- Examine seasonal patterns in monthly sales.
- Include additional factors such as new stores, product prices, customer base, and online sales.
- Compare different time periods, including before and after the introduction of online sales.
- Perform correlation and regression analysis.
- Determine whether advertising remains an important predictor after controlling for other variables.
- Avoid interpreting a high correlation as direct proof of causation.

## CODE :

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Create monthly data for 5 years
np.random.seed(42)

months = pd.date_range(
    start="2021-01-01",
    periods=60,
    freq="MS"
)

time = np.arange(60)

# Advertising expenditure increases over time
advertising = (
    50000
    + time * 1500
    + np.random.normal(0, 4000, 60)
)

# Number of stores increases
```

```
stores = 5 + (time // 12)
```

```
# Product prices gradually increase
```

```
price_index = 100 + time * 0.4
```

```
# Customer base expands
```

```
customers = 10000 + time * 300
```

```
# Online sales introduced in third year
```

```
online_sales = np.where(time >= 24, 1, 0)
```

```
# Seasonal pattern
```

```
seasonality = 15000 * np.sin(
```

```
    2 * np.pi * time / 12
```

```
)
```

```
# Generate sales
```

```
sales = (
```

```
    200000
```

```
    + advertising * 2
```

```
    + stores * 25000
```

```
    + customers * 5
```

```
    + online_sales * 50000
```

```
    + seasonality
```

```
    + np.random.normal(0, 20000, 60)
```

```
)
```

```
# Create DataFrame
```

```
df = pd.DataFrame({  
    "Date": months,  
    "Advertising": advertising,  
    "Sales": sales,  
    "Stores": stores,  
    "Price_Index": price_index,  
    "Customers": customers,  
    "Online_Sales": online_sales  
})
```

```
# Calculate correlation
```

```
correlation = df["Advertising"].corr(df["Sales"])
```

```
print("Correlation between Advertising and Sales:")
```

```
print(correlation)
```

```
# Display first few records
```

```
print(df.head())
```

```
# Line chart
```

```
plt.figure(figsize=(12, 6))
```

```
plt.plot(  
    df["Date"],
```

```
    df["Sales"],
```

```
df["Advertising"],  
label="Advertising Expenditure"  
)
```

```
plt.plot(  
df["Date"],  
df["Sales"],  
label="Sales"  
)
```

```
plt.title("Advertising Expenditure and Sales Over Time")  
plt.xlabel("Year")  
plt.ylabel("Amount")  
plt.legend()  
plt.grid()
```

```
plt.show()
```

```
# Scatter plot
```

```
plt.figure(figsize=(8, 5))
```

```
sns.scatterplot(  
data=df,  
x="Advertising",  
y="Sales"  
)
```

```
plt.title("Advertising Expenditure vs Sales")
```

```
plt.show()
```

```
# Correlation heatmap
```

```
numeric_data = df[
```

```
[
```

```
    "Advertising",
```

```
    "Sales",
```

```
    "Stores",
```

```
    "Price_Index",
```

```
    "Customers",
```

```
    "Online_Sales"
```

```
]
```

```
]
```

```
plt.figure(figsize=(9, 6))
```

```
sns.heatmap(
```

```
    numeric_data.corr(),
```

```
    annot=True,
```

```
    cmap="coolwarm"
```

```
)
```

```
plt.title("Correlation Analysis of Business Variables")
```

```
plt.show()

# Monthly seasonality analysis
df["Month"] = df["Date"].dt.month

monthly_sales = df.groupby("Month")["Sales"].mean()

plt.figure(figsize=(10, 5))

plt.bar(
    monthly_sales.index,
    monthly_sales.values
)

plt.title("Average Monthly Sales Pattern")
plt.xlabel("Month")
plt.ylabel("Average Sales")

plt.show()
```

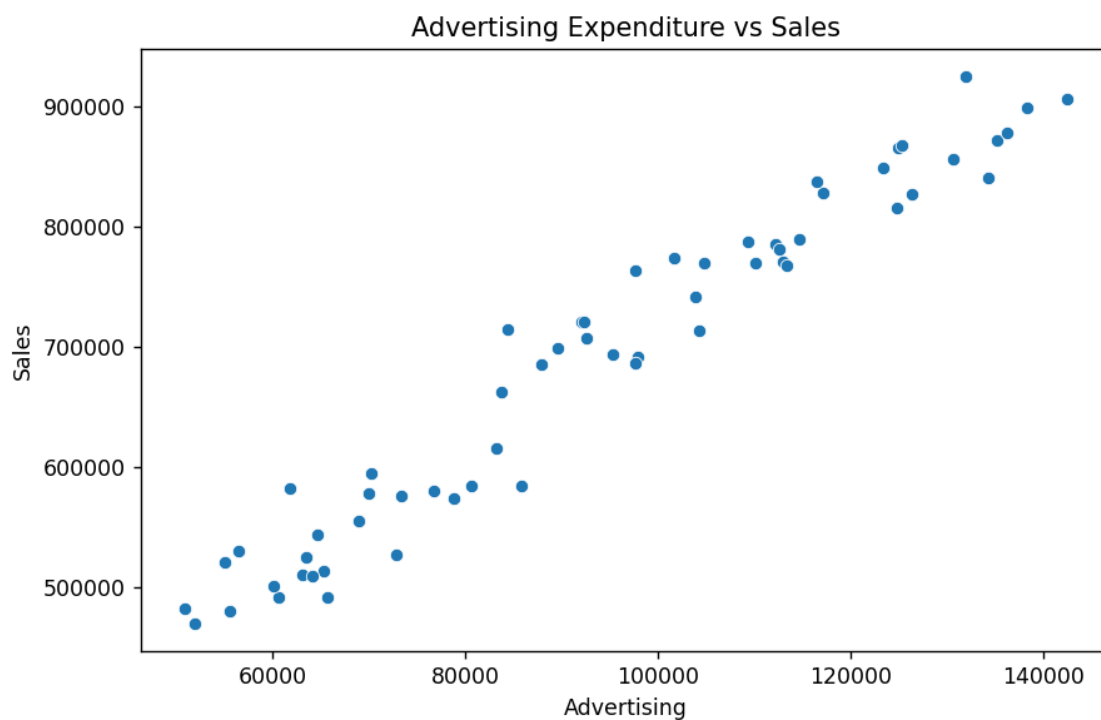
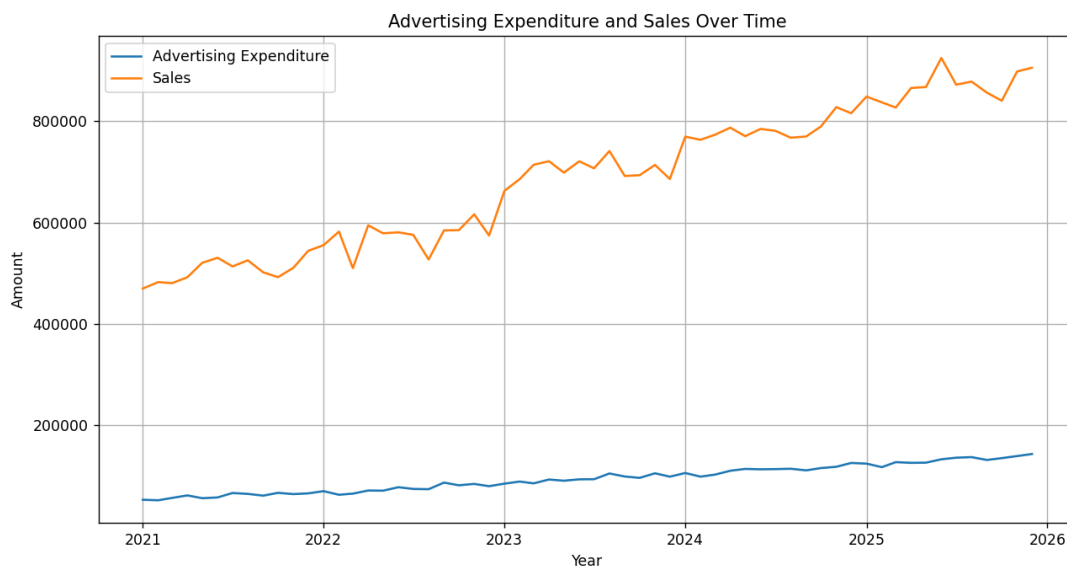
### **Output :**

The analysis may show a strong positive correlation, potentially close to 0.93, between advertising expenditure and sales.

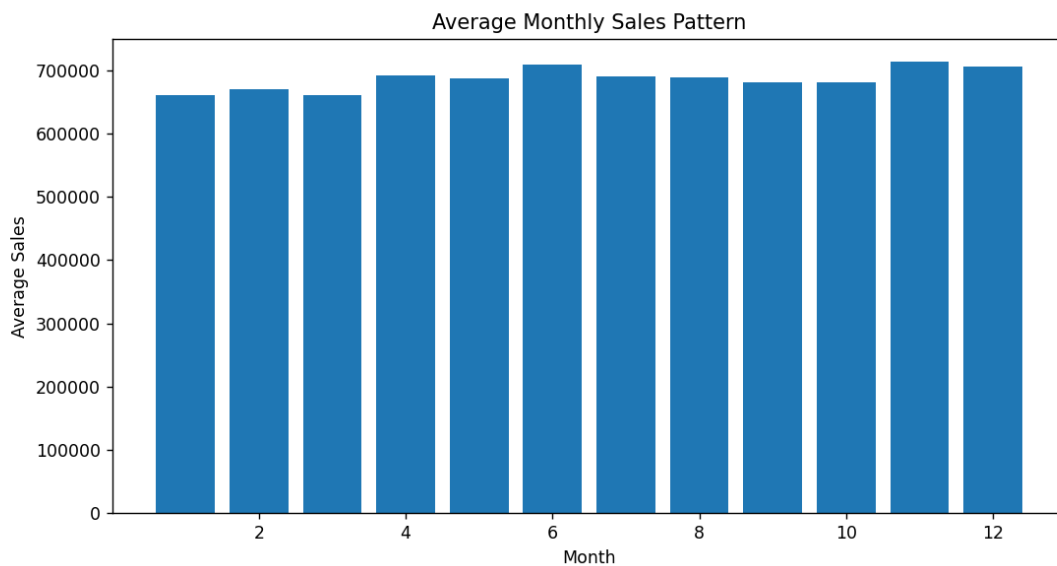
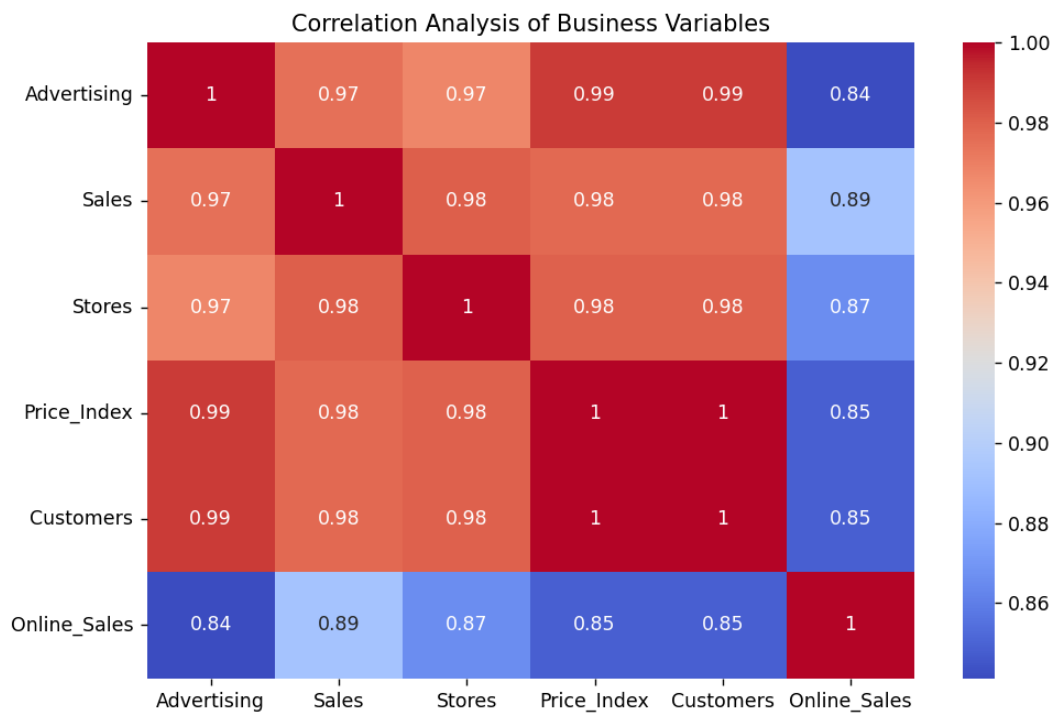
However, the time-series visualization may show that both advertising and sales increase steadily over the five-year period. This means that the strong

correlation could partly result from a common time trend rather than a direct causal relationship.

The heatmap also helps examine relationships between sales and other variables such as:







The monthly analysis can reveal recurring seasonal sales patterns.

## Key Insights :

A correlation of 0.93 represents a strong association between advertising expenditure and sales, but it does not prove that advertising alone caused the increase in sales.

Both variables may have increased over time because of business growth. New stores can increase sales, while an expanded customer base can create additional demand. Product price increases may also increase revenue even if the quantity of products sold remains unchanged.

The introduction of online sales during the third year can create a structural change in sales. Seasonal peaks may also cause repeated increases in revenue during particular months.

Therefore, these factors must be analyzed separately before concluding that increasing advertising expenditure will proportionally increase sales.

## **Conclusion :**

The analysis demonstrates that a strong correlation between advertising expenditure and sales is not sufficient evidence of causation. The relationship may be affected by common time trends, seasonality, store expansion, price changes, customer growth, and the introduction of online sales.

A proper analytical workflow should include visualization, time-series analysis, subgroup comparison, and multiple regression analysis. This allows the company to determine whether advertising has a genuine predictive relationship with sales after controlling for other important variables.

This follows the same analytical principle shown in your sample project: correlation helps identify relationships, but it should be interpreted as an association rather than proof of causation